

Equity Exotics & Structured Products

Interview Preparation Guide

Structuring Desk — Bank of America

Scope: Vanilla options & Greeks · Volatility surface & models · Barrier
& digital options
Autocallable structures · Worst-of / Best-of options · Correlation trading
Structured note construction · Multi-asset modelling

Based on *Exotic Options and Hybrids* (Bouzoubaa & Osseiran)
and practitioner context from the BofA Paris exotic trading desk

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1 The Structured Products Ecosystem

1.1 What is a Structured Product?

A structured product is a pre-packaged investment that combines a **non-risky component** (typically a zero-coupon bond providing capital protection) with a **risky derivative component** (an option or basket of options) to deliver a specific risk/return profile tailored to investor needs.

Anatomy of a Capital-Protected Note

$$\underbrace{\text{Issue Price}}_{\approx 100\%} = \underbrace{P_{\text{ZCB}}(0, T)}_{\text{bond at discount}} + \underbrace{V_{\text{option}}}_{\text{derivative}} + \underbrace{\text{P\&L}}_{\text{bank margin}}$$

where the zero-coupon bond price is $P_{\text{ZCB}} = e^{-rT}$ (continuously compounded) or equivalently $P_{\text{ZCB}} = (1 + r)^{-T}$ in discrete compounding.

Concrete example: 10-year EUR note, rates = 3%, P&L = 0.5%, issue at 98%.

$$P_{\text{ZCB}} = e^{-0.03 \times 10} \approx 74.1\% \quad (\text{guarantees } 100\% \text{ at maturity})$$

$$\text{Available for option} = 98\% - 74.1\% - 0.5\% = 23.4\%$$

This budget buys equity upside exposure. The participation rate is the ratio of available budget to ATM call cost: if a 10-year ATM call on EuroStoxx 50 costs 22%, the participation is $23.4\%/22\% \approx 106\%$ — the investor receives slightly more than the index return.

BofA Context

BofA Paris issues assurance-vie structured notes for French retail clients. These are *very long-dated* (often 8–12 years) because they are placed inside life insurance wrappers for tax advantages. The long maturity has important consequences: (1) the zero-coupon bond is much cheaper at long maturities (more budget for the option), and (2) long-dated options require careful treatment of dividends, forward vol, and interest rate/equity correlation. Swiss clients take shorter-dated structures (2–5 years), often on single stocks.

1.2 Market Participants and Roles

- **Sales:** Contacts clients, understands their risk appetite and regulatory constraints, brings product ideas to structurers.
- **Structurer:** Designs the payoff to meet client objectives within a budget. Prices the structure, constructs the term sheet, and liaises with the trading desk for risk.
- **Exotic Trader:** Hedges the delta and manages the Greeks of the book. Takes the correlation, vol, and skew risks that cannot be fully replicated.
- **Quant:** Builds and maintains pricing models; develops calibration routines and Monte Carlo engines.

Interview Question / Key Insight

Q: Walk me through how you would structure a capital-protected note.

Start with the maturity and rate environment to price the ZCB. The remainder (after P&L) funds the option. Select the option type based on client view: bullish → call or autocall; yield-seeking → worst-of autocall with DI put. Adjust participation by cheapening the option (gearing, averaging, caps, or barriers).

2 Vanilla Options: Foundations

2.1 Payoffs and Put–Call Parity

A European **call** option gives the right to buy the underlying S at strike K at maturity T :

$$C(T) = \max(S_T - K, 0) = (S_T - K)^+$$

A European **put** option gives the right to sell:

$$P(T) = \max(K - S_T, 0) = (K - S_T)^+$$

Moneyness: An option is *at-the-money (ATM)* when $S \approx K$, *in-the-money (ITM)* when exercising immediately has positive value, *out-of-the-money (OTM)* otherwise.

Put–Call Parity (model-free)

$$C(K, T) - P(K, T) = F \cdot e^{-rT} - K \cdot e^{-rT} = e^{-rT}(F - K)$$

where $F = S_0 e^{(r-q)T}$ is the forward price (q = dividend yield). This is model-free: it holds by absence of arbitrage regardless of the price process.

Interview Question / Key Insight

Q: Prove put–call parity without any model.

Consider two portfolios at $t = 0$: (A) long call + Ke^{-rT} cash; (B) long put + long forward. At T : if $S_T > K$, both pay $S_T - K$; if $S_T < K$, both pay 0. By no-arbitrage, $C + Ke^{-rT} = P + Fe^{-rT}$, which rearranges to the formula above.

2.2 Black–Scholes Model

2.2.1 The GBM Assumption and SDE

The Black–Scholes model assumes the asset price S_t follows a *Geometric Brownian Motion* (GBM) under the risk-neutral measure $\mathbb{Q}$:

Black–Scholes SDE (risk-neutral)

$$dS_t = r S_t dt + \sigma S_t dW_t^{\mathbb{Q}}$$

where r is the risk-free rate, σ the (constant) volatility, and $W_t^{\mathbb{Q}}$ a standard Brownian motion under $\mathbb{Q}$.

Applying Itô's lemma to $\ln S_t$ gives:

$$d(\ln S_t) = \left(r - \frac{\sigma^2}{2}\right) dt + \sigma dW_t$$

The term $-\sigma^2/2$ is the *Itô correction*: because the exponential is convex (Jensen's inequality), the log-return drifts below the arithmetic return. This gives the terminal distribution:

$$\ln\left(\frac{S_T}{S_0}\right) \sim \mathcal{N}\left(\left(r - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right)$$

2.2.2 The Black–Scholes PDE

By constructing a delta-hedged portfolio and applying no-arbitrage, the option price $V(S, t)$ must satisfy:

Black–Scholes PDE

$$\underbrace{\frac{\partial V}{\partial t}}_{\Theta} + \underbrace{rS \frac{\partial V}{\partial S}}_{\text{drift term}} + \underbrace{\frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2}}_{\frac{1}{2}\sigma^2 S^2 \Gamma} - rV = 0$$

Equivalently (for a delta-neutral portfolio where $\Delta = 0$): $\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma = rV$.

Interpretation: This PDE encodes the fundamental Gamma/Theta trade-off. A long-option position gains from convexity (positive Γ) but loses time value ($\Theta < 0$). In equilibrium, these exactly offset each other at the risk-free rate.

2.2.3 Black–Scholes Pricing Formula

Solving the PDE with the call payoff as boundary condition yields:

Black–Scholes Formula

$$C_{\text{BS}} = S_0 e^{-qT} N(d_1) - K e^{-rT} N(d_2)$$

$$P_{\text{BS}} = K e^{-rT} N(-d_2) - S_0 e^{-qT} N(-d_1)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

$N(\cdot)$ is the standard normal CDF; q is the continuous dividend yield.

Intuition for $N(d_2)$: Under $\mathbb{Q}$, $N(d_2)$ is the risk-neutral probability that $S_T > K$ (i.e., the call finishes in the money). $N(d_1)$ is a delta-adjusted probability accounting for the fact that the payoff is S_T , not \$1, when exercised.

2.2.4 ATM Approximation — A Key Interview Result**Interview Question / Key Insight**

Q: Give me a quick approximation for the price of an ATM call.

For an ATM forward option ($K = F_T = S_0 e^{(r-q)T}$), $d_1 \approx \sigma\sqrt{T}/2$ and $d_2 \approx -\sigma\sqrt{T}/2$. Since $N'(0) = 1/\sqrt{2\pi} \approx 0.4$:

$$C_{\text{ATM}} \approx 0.4 \cdot \sigma \cdot S_0 \cdot \sqrt{T}$$

Why is this valid? An ATM call is *linear* in σ : its Volga (second derivative in σ) is approximately zero ATM. Therefore price = Vega $\times \sigma$, and Vega_{ATM} $\approx 0.4 \cdot S_0 \cdot \sqrt{T}$.

Example: 1-year ATM call, $S_0 = 100$, $\sigma = 20\%$: $C \approx 0.4 \times 0.20 \times 100 \times 1 = 8\%$. Black–Scholes exact: $\approx 7.97\%$. Very accurate.

3 Option Greeks: Sensitivities and Their Meaning

Greeks quantify how the option price changes with each market parameter. They are the primary risk management language of any derivatives desk.

3.1 Delta (Δ)

Delta

$$\Delta_{\text{call}} = e^{-qT} N(d_1), \quad \Delta_{\text{put}} = -e^{-qT} N(-d_1)$$

Range: $\Delta_{\text{call}} \in [0, 1]$, $\Delta_{\text{put}} \in [-1, 0]$.

Δ is the number of units of the underlying needed to hedge the option. For a sold call with $\Delta = 0.6$, the trader must **buy** 0.6 units of the underlying.

Intuition: For a deep ITM call, $\Delta \rightarrow 1$ (the option behaves like the stock); for a deep OTM call, $\Delta \rightarrow 0$ (very unlikely to exercise).

Effect of volatility on Δ : Higher vol diffuses the option's probability of exercise across a wider range of spot levels. OTM options get a higher Δ (more likely to expire ITM) and ITM options get a lower Δ . Geometrically, the S-curve flattens toward $\Delta \equiv 0.5$ for all strikes as $\sigma \rightarrow \infty$. Low vol makes Δ a sharp step function concentrated near the strike.

Effect of interest rates on Δ : Higher r raises the forward $F = Se^{(r-q)T}$, which increases d_1 , raising Δ_{call} and decreasing $|\Delta_{\text{put}}|$.

Call Delta vs. Spot – effect of volatility ($T = 1\text{Y}$, $r = 0$, $q = 0$)

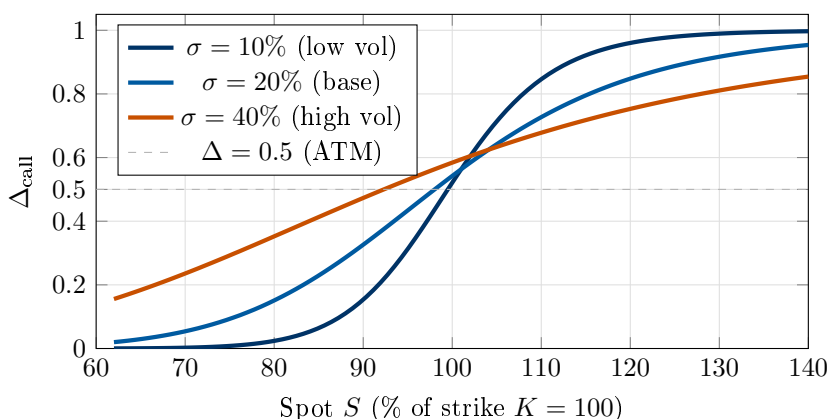


Figure 1: Call Δ vs. spot for three volatility levels. Low vol: sharp S-curve near $K = 100$. High vol: flat curve; all options approach $\Delta = 0.5$. At $S = K$, $\Delta \approx 0.5$ regardless of vol.

Interview Question / Key Insight

Q: If vol goes down, and you are long OTM calls, do you buy or sell the underlying?
 Lower vol $\Rightarrow$ lower Δ for OTM calls. Long calls $\Rightarrow$ portfolio Δ decreases $\Rightarrow$ you need to **buy** the underlying to rebalance to delta-neutral.

3.2 Gamma (Γ)

Gamma

$$\Gamma = \frac{N'(d_1) e^{-qT}}{S_0 \sigma \sqrt{T}}$$

where $N'(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ is the standard normal PDF. Gamma is the same for calls and puts (by put-call parity).

Γ is the *convexity* of the option price in S . It is largest ATM and short-dated. Gamma hedging requires trading other options (since Δ -hedges with the underlying add no convexity).

Effect of volatility on Γ : Higher vol spreads the Gamma peak across a wider range of spot levels (the bell curve flattens and widens). The peak value at ATM *decreases* because $\Gamma_{\text{ATM}} \propto 1/\sigma$: more vol dilutes the convexity per unit spot move. Low vol concentrates a very sharp Gamma spike near the strike — this creates severe re-hedging risk for near-expiry options.

Effect of rates on Γ : Rates have a minor effect through e^{-qT} and the forward level. Higher dividends lower the forward, shifting the Gamma peak slightly downward in spot.

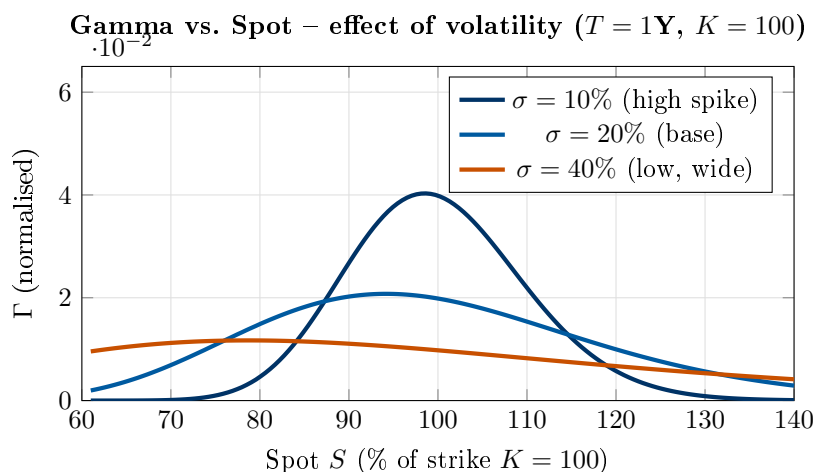


Figure 2: Gamma vs. spot. High vol $\rightarrow$ low wide bell (Gamma diluted). Low vol $\rightarrow$ tall narrow spike at the strike. Near expiry, the spike becomes even sharper and can cause severe re-hedging costs.

3.3 The Gamma–Theta Trade-off

This is perhaps the most fundamental concept in options trading. From the BS PDE with $\Delta = 0$:

P&L of a Delta-Hedged Position

$$\delta\Pi = \Theta \cdot \delta t + \frac{1}{2} \Gamma \cdot (\delta S)^2$$

If realized vol equals implied vol, the two terms exactly cancel. The breakeven daily move is:

$$|\delta S|^* = \sigma_{\text{impl}} S \sqrt{\delta t} = \frac{\sigma_{\text{impl}} \cdot S}{\sqrt{252}}$$

Key Intuition

Long option (long Γ , short Θ): You pay time decay every day. You *profit* only if realized vol exceeds the implied vol you paid for. Each day you do not get large moves, you lose money.

Short option (short Γ , long Θ): You collect premium (time decay) daily. You *lose* if the underlying moves more than the breakeven level $\sigma_{\text{impl}}/\sqrt{252}$ per day.

The P&L of a delta-hedged book over a day is:

$$\text{Daily P\&L} = -\Theta \cdot \delta t + \frac{1}{2} \Gamma \cdot (\delta S_{\text{realized}})^2 = \frac{1}{2} \Gamma S^2 (\sigma_{\text{realized}}^2 - \sigma_{\text{implied}}^2) \delta t$$

This is why option traders care deeply about implied vs. realized vol.

Interview Question / Key Insight

Q: You have a delta-neutral book with daily Theta of $-\$1,000$ and you know Gamma = 5,000. The stock is at $\$100$ and has 16% annual vol. If the stock moves $\pm 2\%$ today, what is your P&L?

Daily breakeven: $\sigma_{1d} = 16\%/\sqrt{252} = 1\%$. Move is 2%.

$$\text{P\&L} = -1,000 + \frac{1}{2} \times 5,000 \times (2\%)^2 \times (100)^2/100$$

Alternatively using the formula $\frac{1}{2}\Gamma(\delta S)^2$: $\frac{1}{2} \times 5,000 \times (2)^2 = \$10,000$ in gamma P&L, minus $\$1,000$ theta = $+\$9,000$.

3.4 Vega ($\mathcal{V}$)**Vega**

$$\mathcal{V} = S_0 e^{-qT} N'(d_1) \sqrt{T}$$

Vega is identical for calls and puts. It is maximized ATM and increases with $\sqrt{T}$.

Vega measures the sensitivity to a 1% (i.e. 1 vol point) change in implied volatility. Vega is bucketed per maturity: a complex book will have Vega in each tenor from 1M to 10Y.

Effect of volatility on Vega: For OTM/ITM options, Vega itself depends on vol (this is Volga). An ATM option's Vega is nearly constant in σ (Volga ≈ 0). For OTM options, Vega increases as σ rises (the option becomes more sensitive to vol changes).

Effect of rates on Vega: Higher rates lower the forward slightly (through discounting), shifting the peak of Vega. Since $\text{Vega} = S e^{-qT} N'(d_1) \sqrt{T}$, higher q (dividend yield) reduces Vega proportionally via the e^{-qT} factor.

ATM call is linear in vol. Because $\text{Volga}_{\text{ATM}} \approx 0$, an ATM call has $\partial^2 C / \partial \sigma^2 \approx 0$, so $C \approx \mathcal{V}_{\text{ATM}} \times \sigma$ (this justifies the approximation formula above).

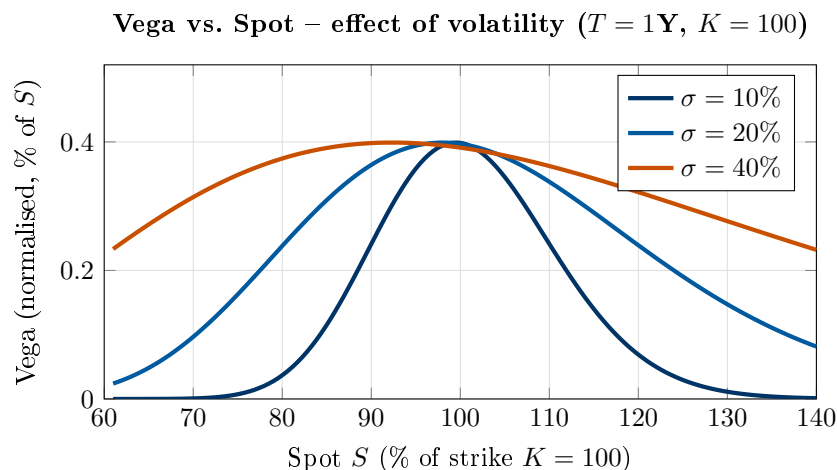


Figure 3: Vega vs. spot. Vega peaks ATM for all vol levels (same bell shape as Gamma, but scaled by $\sigma\sqrt{T}$ vs $1/(\sigma\sqrt{T})$). Higher vol flattens and widens the Vega profile. OTM/ITM options have lower Vega but become more Vega-sensitive as vol rises (Volga effect).

3.5 Theta (Θ)

Theta (for a call, zero rates simplified)

$$\Theta = -\frac{S_0 N'(d_1) \sigma}{2\sqrt{T}}$$

Theta is always negative for long options (options lose value as time passes, all else equal).

Effect of volatility on Theta: Since $\Theta \propto \sigma$, higher vol means more time decay per day — an option with more implied vol has more time value to bleed. The relationship is essentially linear at ATM.

Time effect: As $T \rightarrow 0$, $|\Theta| \propto 1/\sqrt{T} \rightarrow +\infty$ for ATM options. The ATM Theta accelerates sharply near expiry. OTM options conversely lose nearly zero Theta near expiry (they are already close to worthless). This asymmetry is important to understand: do not confuse Theta being small (OTM, near expiry) with the option having no risk — it simply has no time value left to lose.

Theta and Gamma are linked by the BS PDE. Long Gamma $\Leftrightarrow$ short Theta. Near expiry, Theta accelerates (especially for ATM options whose Γ spikes).

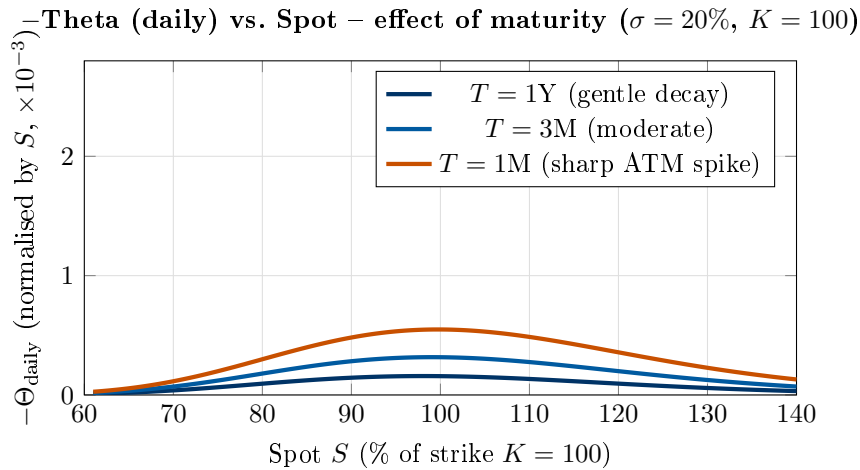


Figure 4: Daily Theta (absolute, normalised) vs. spot for $\sigma = 20\%$, three maturities. ATM options decay fastest, and the rate accelerates near expiry. OTM options always decay more slowly and their Theta peak disappears as they expire worthless.

3.6 Rho (ρ)

$\rho = \partial C / \partial r$. For a long call: higher rates $\rightarrow$ higher forward $\rightarrow$ call worth more. For a long put: higher rates $\rightarrow$ lower present value of strike and lower forward $\rightarrow$ put worth less.

Practical note: For long-dated structures (8–12 years), Rho is material and must be carefully managed through the interest rate desk.

3.7 Volga and Vanna

Second-order Greeks

$$\text{Volga} = \frac{\partial^2 V}{\partial \sigma^2} = \frac{\partial \mathcal{V}}{\partial \sigma} \quad (\text{curvature in vol})$$

$$\text{Vanna} = \frac{\partial^2 V}{\partial S \partial \sigma} = \frac{\partial \Delta}{\partial \sigma} = \frac{\partial \mathcal{V}}{\partial S} \quad (\text{cross-sensitivity: spot and vol})$$

Volga is relevant for OTM/ITM options: they are convex in vol (if vol rises a lot, their vega rises). ATM options have near-zero Volga. Structures with high Volga (cliquets, Napoleons) require stochastic volatility models.

Vanna tells us how much our delta hedge must be adjusted when vol moves. Important for barrier options and any structure with significant skew sensitivity.

3.8 Summary of Greek Positions

Position	Δ	Γ	$\mathcal{V}$	Θ	ρ
Long call	+	+	+	−	+
Short call	−	−	−	+	−
Long put	−	+	+	−	−
Short put	+	−	−	+	+

4 Volatility Surface, Skew, and Term Structure

4.1 Implied Volatility and the Surface

The *implied volatility* $\sigma_{\text{impl}}(K, T)$ is the value of σ that, plugged into the BS formula, returns the market price of the option with strike K and maturity T . It is a market-observable quantity, not a model prediction.

The volatility surface $\sigma(K, T)$ plots implied vol across strikes (skew) and maturities (term structure). Under Black–Scholes with constant σ , the surface would be flat. In reality it is not, which tells us that log-returns are not normally distributed.

4.2 The Skew

For equity indices, OTM puts are more expensive than OTM calls at the same distance from ATM. This gives the classic *downward-sloping skew*: $\sigma(K)$ decreases as K increases.

Key Intuition

Why does equity skew exist?

1. **Fat left tail / crash risk:** Equity returns have larger downside moves than a normal distribution predicts. OTM puts protect against crashes and carry a premium.
2. **Supply/demand:** Institutional investors systematically buy puts to hedge equity portfolios, driving up OTM put prices.
3. **Leverage effect:** As the stock falls, firm leverage increases, which increases volatility. This creates a negative correlation between spot and vol.

Skew and its effect on digital pricing

A digital call paying \$1 if $S_T \geq K$ can be replicated as a call spread:

$$\text{Digital}(K) = -\frac{\partial C(K, \sigma(K))}{\partial K} = N(d_2) + \underbrace{\mathcal{V} \times \left| \frac{\partial \sigma}{\partial K} \right|}_{\text{skew correction}}$$

Skew makes digitals *more expensive* when the seller is short them (skew correction is positive).

4.3 Term Structure of Volatility

Implied vol also varies with maturity T . Short-dated vol ($< 3M$) reacts sharply to near-term events; long-dated vol ($> 2Y$) is more stable and mean-reverting. The term structure can be in contango (vol increasing with T , typical in calm markets) or backwardation (short-dated vol high, e.g. during a crisis).

Key consequence for long-dated notes: BofA Paris issues 10-year structures, so the relevant vol is the 10-year implied vol, which is significantly different from short-dated vol and harder to hedge (fewer liquid instruments).

5 Volatility Models

Understanding vol models matters because they determine how exotic option Greeks are computed and how risk is managed.

5.1 Black–Scholes: Flat Volatility

BS assumes constant σ . It fails to capture skew or term structure. Despite this, BS is the universal quoting convention: even complex exotic options are often quoted in terms of the implied vol that a BS formula would need to match market prices. It is the language, not the model.

When is BS sufficient? For plain vanilla European options on a specific strike (each gets its own $\sigma_{\text{impl}}(K, T)$), BS is fine. For path-dependent or multi-asset options, BS breaks down.

5.2 Local Volatility

Local volatility extends BS by making the diffusion coefficient a *deterministic function* of both spot and time:

Local Volatility SDE

$$dS_t = r S_t dt + \sigma(S_t, t) S_t dW_t$$

The local vol function is extracted from market prices via Dupire's formula:

$$\sigma^2(K, T) = \frac{2 \frac{\partial C}{\partial T} + (r - q) K \frac{\partial C}{\partial K} + q C}{K^2 \frac{\partial^2 C}{\partial K^2}}$$

Key properties:

- Still *one-factor* model (single Brownian motion); no additional randomness.
- Calibrates *exactly* to the entire vol surface — this is its main appeal.
- Used for most path-dependent single-asset exotics (barrier options, autocalls).

Calibration types:

- **Exact date fitting:** Focus calibration on one maturity (for payoffs that only depend on S_T , e.g. a European-barrier monitored only at expiry).
- **Smooth surface:** Calibrate across all maturities (for continuously-monitored barriers, which have vega sensitivity at all times up to T).

Limitation — forward skew: Local vol models imply that forward skew *flattens* over time. Intuitively: $\sigma(S_t, t)$ is deterministic, so as the model simulates forward, the conditional distribution narrows. This causes cliquets and forward-starting options to be *mispriced* if local vol is used naively.

Caution / Common Mistake

If you use a local vol model to price a cliquet or any product sensitive to forward skew, you will almost certainly underprice the risk. These products require stochastic volatility.

5.3 Stochastic Volatility (Heston)

In stochastic vol models, volatility is itself a random process:

Heston Model

$$\begin{aligned} dS_t &= r S_t dt + \sqrt{v_t} S_t dW_t^S \\ dv_t &= \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v \\ dW_t^S \cdot dW_t^v &= \rho dt \end{aligned}$$

where: v_t = instantaneous variance, κ = mean reversion speed, θ = long-run variance, ξ = vol of vol, ρ = spot-vol correlation.

Heston generates implied skew (through ρ) and vol convexity (through ξ , the vol-of-vol). It has semi-closed-form solutions for European options, making calibration tractable.

When is stochastic vol necessary?

- Products with **Vega convexity** (Volga $\neq 0$): cliquets, Napoleons, variance swaps.
- Products sensitive to **forward skew**: all forward-starting structures.
- When the market's implied vol moves significantly and the product's Vega changes with it (non-linear vol exposure).

5.4 SABR Model

SABR is a stochastic vol model designed to capture correct smile dynamics, particularly used in interest rate markets:

$$dF_t = \alpha_t F_t^\beta dW_t, \quad d\alpha_t = \nu \alpha_t dZ_t, \quad \langle dW_t, dZ_t \rangle = \rho dt$$

SABR has an approximate closed-form expression for implied vol, making calibration simple. It is used in copula-based multi-asset pricing as the marginal distribution model.

6 Barrier Options

6.1 Payoff Mechanisms

Barrier options are path-dependent: their payoff depends on whether S_t breaches a trigger H during $[0, T]$.

Knock-out and Knock-in Parity

$$\text{KO}(K, T, H) + \text{KI}(K, T, H) = \text{Vanilla}(K, T)$$

A knock-out option dies if S touches H ; a knock-in activates only if S touches H .

Taxonomy:

- **Down-and-Out (DO)**: Barrier below spot; option dies if S falls to H .
- **Down-and-In (DI)**: Option activates if S falls to H .
 $\hookrightarrow$ *The down-and-in put is the most important for structured products.*
- **Up-and-Out (UO)**: Barrier above spot; option dies if S rises to H .
- **Up-and-In (UI)**: Option activates if S rises to H .

6.2 The Down-and-In Put: Structured Products Workhorse

The DI put is the key risk component in autocall + DI put structures. Its payoff is:

$$\text{DI Put} = (K - S_T)^+ \cdot \mathbf{1}_{\{\min_{t \in [0, T]} S_t \leq H\}}$$

The investor effectively *sells* this put to the bank in exchange for a higher coupon.

Sensitivities of the DI put buyer (bank):

Parameter	DI Put Buyer (Bank)	Direction
Interest rates (forward)	Short forward $\rightarrow$ short rates	—
Dividends	Long dividends	+
Borrow costs	Long borrow costs	+
Volatility	Long vol (higher vol $\rightarrow$ more knock-in prob.)	+
Skew	Long skew (downside vol is higher)	+
Barrier level H	Long barrier (higher H = easier to KI)	+

Near-barrier risk: As $S_t \rightarrow H$, the option's Δ and Γ become very large and unstable. A small move through H causes a discontinuous change in value. This is managed via a **barrier shift**:

Discrete Barrier Shift (Broadie–Glasserman–Kou)

$$H' = H \cdot e^{\pm 0.5826 \sigma \sqrt{T/m}}$$

where m = number of monitoring dates, + for short barrier, — for long barrier.

The shift essentially compensates for the difference between continuously and discretely monitored barriers, and also accounts for liquidity (the digital size around the barrier).

6.3 Near-Barrier Complexity: Greeks and the Pin Risk

The most treacherous aspect of barrier options is their behavior when the spot approaches the barrier level H . The Greeks become extreme and can change sign, creating severe hedging challenges.

Up-and-Out Call (UO Call) near the barrier: Consider a call struck at $K = 100\%$ knocking out at $H = 120\%$. When $S \lesssim H$:

- The option's value is approaching zero (it is about to die).
- But a vanilla call at the same spot would be worth $\approx 20\%$ (deep ITM).
- Therefore the *value drops sharply* as $S \rightarrow H$: the option is worth nothing at the barrier.

This creates a *negative Delta* near the barrier for the UO call: as spot rises, the option loses value (it is about to knock out). The Delta switches sign from positive (far from barrier) to negative (near barrier). Similarly, Gamma can be large and negative near the barrier.

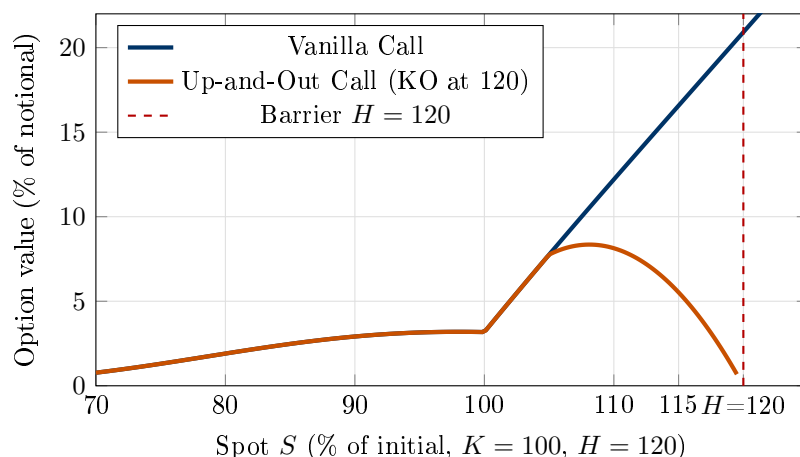
Up-and-Out Call vs. Vanilla Call ($K = 100$, $H = 120$, $\sigma = 20\%$, $T = 1Y$)

Figure 5: The UO call drops to zero at the barrier while the vanilla call would be worth $\approx 20\%$. Near the barrier, the UO call's Delta becomes negative (option loses value as spot rises). This creates a dangerous hedging situation: the trader must *sell* the underlying as it rallies.

The pin risk (digital risk at the barrier): At expiry, if S_T is exactly at H , the option jumps discontinuously from its vanilla value to zero. The Delta is theoretically $-\infty$. In practice, this creates a huge exposure to a small move through the barrier. This is exactly the “digital” at the barrier: the barrier shift technique (Section 6) is designed to manage this by pricing as if the barrier were slightly further away.

Down-and-In Put near the barrier: As $S \rightarrow H$ from above:

- The DI put's value rises sharply (it is about to activate and become a valuable vanilla put).
- Delta becomes very negative (long the put means short spot).
- Gamma spikes to very large positive values (the option's delta changes rapidly).

In a structured product where the bank is long the DI put, a sudden breach of the barrier means the bank suddenly becomes short a large delta position — the trader must rapidly buy a large quantity of shares to rebalance. The volume required can exceed daily liquidity, which is why the barrier shift is also a *liquidity-based* adjustment.

Barrier options have important skew sensitivity. A DI put that activates on the downside is priced at a volatility level corresponding to the barrier (which is OTM, hence higher implied vol due to skew):

Key Intuition

A 60% DI put on a stock with ATM vol of 20% but 30% vol at the 60% strike will be significantly mispriced if one uses flat vol. The skew benefit to the buyer of the DI put ranges from 70% to 100% of the skew correction (100% is aggressive, 70% is conservative). Correct calibration requires a model fitted to the downside skew — typically local vol with exact date fitting for European barriers.

7 Digital Options

7.1 Payoff and Replication

A digital call (also called binary or cash-or-nothing) pays a fixed amount X if $S_T \geq K$:

$$\text{Digital Call payoff} = X \cdot \mathbf{1}_{\{S_T \geq K\}}$$

Digital as a Call Spread

$$\text{Digital}(K) = \lim_{\epsilon \rightarrow 0} \frac{C(K - \epsilon) - C(K + \epsilon)}{2\epsilon} = -\frac{\partial C}{\partial K}$$

In practice: replicate with a tight call spread: buy call at $K - \epsilon$, sell call at $K + \epsilon$. The digital price is approximately $\frac{C(K - \epsilon) - C(K + \epsilon)}{2\epsilon}$.

Under Black–Scholes: Digital Call = $e^{-rT} N(d_2)$.

Skew effect on digitals:

$$\text{Digital}(K) = N(d_2) + \mathcal{V} \times \underbrace{\left| \frac{\partial \sigma}{\partial K} \right|}_{\text{skew steepness}}$$

In the presence of a downward skew ($\partial \sigma / \partial K < 0$), selling a digital call (i.e. shorting a call spread) means selling the higher-vol option and buying the lower-vol option. The seller is effectively **short skew**.

7.2 Greeks of Digitals

Vega: A digital's Vega is positive when the forward is below the trigger (higher vol = higher probability of reaching K) and negative when the forward is above the trigger. This is the opposite of vanilla options, where Vega is always positive for long positions.

Gamma: Near the barrier, the digital has very large (and potentially negative) Gamma — its Delta can change sign. This requires careful Greek management and is why barrier shifts are applied.

Interview Question / Key Insight

Q: You sell a digital call at 110%. The stock forward is 100%. What is your Vega position?

Forward (100%) is below the trigger (110%). Higher vol $\rightarrow$ higher probability of reaching trigger $\rightarrow$ digital becomes more expensive $\rightarrow$ you (the seller) are **long Vega**. This is counter-intuitive: selling the digital means you want vol to go up (so you can buy it back cheaper... wait, you have sold it, so you benefit if vol goes up and makes it more expensive? No — you are long Vega means you benefit from vol going up). Let us be precise: *long the digital = long Vega when forward below trigger; short the digital = short Vega in the same scenario*. Selling a digital where the forward is below the trigger: you are short Vega (you lose money if vol goes up, because your short position in the digital becomes worth more).

8 Autocallable Structures

8.1 Structure and Payoff

An autocallable (auto-trigger) note automatically redeems early if the underlying crosses a specified trigger on predetermined observation dates. The standard single-asset structure:

Autocall Payoff — Single Asset

At each observation date t_i ($i = 1, \dots, n$):

$$\text{Coupon}(t_i) = \text{Notional} \times (1 + i \cdot C) \times \mathbf{1}_{\{S(t_i)/S_0 \geq H_{\text{auto}}\}} \times \mathbf{1}_{\{\text{not previously autocalled}\}}$$

If the note reaches maturity without autocalling, the investor receives par (capital protected) or par + a participation feature (e.g. APN). If a DI put has been sold, capital may be at risk.

Step-up triggers are common: the autocall trigger increases each year (e.g. 100%, 105%, 110% in years 1, 2, 3). This structure is cheaper (lower digital prices) because the step-up decreases the probability of autocalling in later years.

BofA Context

BofA Paris autocalls on indices (EuroStoxx 50, S&P 500, Nikkei) for assurance-vie:

- Typical maturity: 8–12 years with annual or semi-annual observation dates.
- Long dated because of tax advantages of assurance-vie wrappers.
- 10-year autocalls expose the bank to dividend risk and long-dated vol — both are harder to hedge than for short-dated structures.

8.2 Decomposition as a Strip of Path-Dependent Digitals

An autocall is equivalent to a series of path-dependent digital options. The coupon at t_i is a digital that pays only if (a) the trigger is hit at t_i AND (b) it was not already hit at $t_1, \dots, t_{i-1}$.

Undiscounted conditional probabilities: A quick sanity check when pricing is to compute the undiscounted probability of each coupon being paid:

$$p_i = P\left[S(t_i) \geq H_i \mid S(t_j) < H_j \forall j < i\right]$$

These probabilities decrease over time (if the note did not autocall in year 1, it is less likely to in year 2). The price of each autocall digital is $p_i \times e^{-rt_i} \times \text{Coupon}_i$.

Pricing key fact: Autocall structures tend to either autocall in year 1, or survive close to maturity. The distribution of autocall times is bimodal.

8.3 Risk Analysis of Single-Asset Autocall

The seller of an autocallable (the bank):

Risk Factor	Position	Reason
Forward / Interest rates	Short	Higher forward $\rightarrow$ autocall more likely $\rightarrow$ fewer coupons
Dividends	Long	Higher div $\rightarrow$ lower forward $\rightarrow$ longer-lived autocall
Borrow costs	Long	Higher borrow $\rightarrow$ lower forward
Volatility	Depends	If forward < trigger: long vol; if forward > trigger: short vol
Skew	Short	Seller is short a digital: pays more expensive due to skew
Trigger level	Long	Higher trigger $\rightarrow$ autocall later $\rightarrow$ seller collects more time

Vega position: Because the autocall is a strip of digitals, the Vega depends on where the forward sits relative to each trigger. At inception, with the forward typically near or below the first-year trigger (set at 100%), the seller is usually *long Vega*.

8.4 Autocall with Down-and-In Put

Adding a DI put feature allows the client to receive a higher coupon in exchange for taking downside risk. The full structure (sold by bank, i.e. the bank is long everything):

Autocall + DI Put Structure (Bank's book)

Bank position = Long autocall digitals + Long DI put

At observation dates: bank pays coupon only if autocall event occurs.

At maturity: bank pays notional if all underlyings above KI barrier; if any below, bank delivers shares (physical delivery) at strike price.

Skew position complexity: Short autocall (bank sells structure) → short the digitals → short skew. Long DI put → long skew. The net skew position must be computed numerically; neither direction dominates a priori and it depends on the specific parameters.

8.5 Interest Rate / Equity Correlation

The autocall's redemption time is random and equity-dependent. This creates exposure to the correlation between interest rates and the equity underlying.

Key Intuition

Positive equity/rate correlation: When equity rises (autocall more likely), rates also rise (by assumption). The bank (long autocall, short interest rate risk) needs to rebalance its IR hedge: sell 2-year bonds, buy 1-year bonds. But rising rates mean bond prices fall, and the bank is selling the bond that fell more (2-year). This creates a loss: the seller of the autocall should charge more if equity/rate correlation is positive.

Negative correlation: The opposite hedge rebalancing generates profits → the autocall should be cheaper for the bank.

In practice, this correlation is priced as a small margin (a few basis points) unless a full hybrid stochastic rates/equity model is used.

9 Basket Options

9.1 Basket Variance and Implied Correlation

A basket $W = \sum_i w_i S_i$ has a variance driven by the full covariance structure of its components:

Basket Variance — n Assets

$$\sigma_{\text{basket}}^2 = \sum_{i=1}^n w_i^2 \sigma_i^2 + 2 \sum_{i < j} w_i w_j \sigma_i \sigma_j \rho_{ij}$$

Equivalently in matrix form: $\sigma_{\text{basket}}^2 = \mathbf{w}^\top \Sigma \mathbf{w}$, where $\Sigma_{ij} = \sigma_i \sigma_j \rho_{ij}$.

Extracting implied correlation from option prices: given the implied vol of the basket option $\sigma_{\text{basket}}^{\text{impl}}$ (quoted, or backed out from a basket option price via Black–Scholes) and the implied vols σ_i^{impl} of each component (from liquid vanilla options), invert the variance formula for a single **average implied correlation** $\bar{\rho}$:

Average Implied Correlation

$$\bar{\rho} = \frac{\left(\sigma_{\text{basket}}^{\text{impl}}\right)^2 - \sum_i w_i^2 \left(\sigma_i^{\text{impl}}\right)^2}{2 \sum_{i < j} w_i w_j \sigma_i^{\text{impl}} \sigma_j^{\text{impl}}}$$

Equipweighted, equal-vol special case ($w_i = 1/n$, $\sigma_i = \sigma$):

$$\bar{\rho} = \frac{n \left(\sigma_{\text{basket}}^{\text{impl}} / \sigma\right)^2 - 1}{n - 1}$$

This is the standard **implied correlation** measure used in dispersion trading desks — it answers: “if all pairwise correlations were equal to a single number, what would that number be to reproduce the observed basket vol?”

9.2 Pricing the Implied Correlation of a Custom Sub-Basket

In practice, a client may want a basket option on a custom sub-index (e.g. 5 banking stocks) for which no liquid basket option is quoted. The standard technique borrows the implied correlation of a related liquid index.

The λ Parametrization

$$\rho_{\text{implied}}^{\text{index}} = \rho_{\text{realized}}^{\text{index}} + \lambda (1 - \rho_{\text{realized}}^{\text{index}})$$

λ measures the **correlation risk premium**: the fraction of the gap between realized correlation and perfect correlation (1) that the market prices in. Re-apply the *same* λ to any pair (or sub-basket) within the same index:

$$\rho_{\text{implied}}^{AB} = \rho_{\text{realized}}^{AB} + \lambda (1 - \rho_{\text{realized}}^{AB})$$

Procedure:

1. Compute $\rho_{\text{realized}}^{\text{index}}$ from historical component returns (via the basket variance formula, solved for $\bar{\rho}$ using realized vols).
2. Compute $\rho_{\text{implied}}^{\text{index}}$ from the liquid index option’s implied vol vs. component implied vols (formula above).
3. Solve for $\lambda = \frac{\rho_{\text{implied}}^{\text{index}} - \rho_{\text{realized}}^{\text{index}}}{1 - \rho_{\text{realized}}^{\text{index}}}$.
4. For the custom sub-basket: compute ρ_{realized} for each pair (or the whole sub-basket) from historical data, then apply the same λ to obtain ρ_{implied} .

Key Intuition

This is a linear interpolation between ρ_{realized} and 1, with λ as the weight on the “perfect correlation” scenario. It is bounded ($\rho_{\text{implied}} \in [\rho_{\text{realized}}, 1]$) by construction, unlike a naive ratio $\rho_{\text{realized}} \times (\rho_{\text{impl}}^{\text{index}} / \rho_{\text{real}}^{\text{index}})$ which can exceed 1. The key assumption is that λ is a property of the index/regime, not of the specific pair — a single number transfers the correlation premium across all pairs.

9.3 Why Basket Vega Is Sensitive to Correlation, and Why It Is a Delta-Like Effect

A vanilla call on a basket has the usual positive Vega with respect to the *basket's own* volatility. But correlation does not enter through a separate “correlation Vega” — it enters *through* the basket Vega, via the chain rule.

Correlation Sensitivity via Vega

$$\frac{\partial C_{\text{basket}}}{\partial \rho_{ij}} = \underbrace{\frac{\partial C_{\text{basket}}}{\partial \sigma_{\text{basket}}}}_{\mathcal{V}_{\text{basket}} > 0} \times \frac{\partial \sigma_{\text{basket}}}{\partial \rho_{ij}}$$

The subtlety: basket *variance* is linear in ρ_{ij} :

$$\frac{\partial \sigma_{\text{basket}}^2}{\partial \rho_{ij}} = 2w_i w_j \sigma_i \sigma_j \quad (\text{constant})$$

but basket *volatility* is the square root, so by the chain rule:

$$\frac{\partial \sigma_{\text{basket}}}{\partial \rho_{ij}} = \frac{w_i w_j \sigma_i \sigma_j}{\sigma_{\text{basket}}}$$

This is **decreasing in** ρ_{ij} (since σ_{basket} grows with ρ_{ij} in the denominator): the marginal sensitivity of basket vol to correlation shrinks as correlation is already high. The variance is linear in ρ ; the volatility (and hence the price) is concave in ρ .

Key Intuition

A call on a basket is **long correlation** (since $\mathcal{V}_{\text{basket}} > 0$ and $\partial \sigma_{\text{basket}} / \partial \rho_{ij} > 0$). A vendor of a basket call is therefore **short correlation** — equivalently, short the covariance between the components: $\partial \sigma_{\text{basket}}^2 / \partial \text{Cov}(i, j) = 2w_i w_j > 0$ always. This holds for both basket calls and puts (Vega has the same sign for both, by put–call parity).

9.4 Pricing Method: Vanilla BS vs. Full Multi-Asset Monte Carlo

	a) BS + Index Proxy	b) Multi-Asset Monte Carlo
Treats basket as	Single log-normal asset	n correlated underlyings
Vol input	$\sigma_{\text{basket}}^{\text{impl}}(K, T)$ directly, or a liquid index/sector proxy if untraded	Local-vol or LSV surface <i>per asset</i> , calibrated individually
Correlation	Implicit, embedded in σ_{basket}	Explicit, simulated via Cholesky/copula
Captures component skew?	No — distribution of the basket is approximated as log-normal (moment matching, Brigo et al.)	Yes — full component skew preserved
When to use	Vanilla payoff, no skew sensitivity beyond basket-level ATM/skew	Payoff sensitive to <i>which</i> component drives the outcome, or requires correct tail behaviour

Caution / Common Mistake

A weighted sum of log-normal assets is **not itself log-normal**. Approach (a) matches only the first two moments (mean, variance) of the true basket distribution to a log-normal proxy — higher moments (skew, kurtosis of the basket distribution) are lost. This is acceptable for ATM vanilla baskets but breaks down for OTM strikes or any payoff sensitive to the tails.

BofA Context

When no liquid options exist on the basket's components (e.g. illiquid mid-caps), a common proxy is to borrow the skew of a related liquid sector index, scaled down conservatively (e.g. 75% of the index skew) to reflect the basket's own diversification. The Vega hedge ratio $\beta = \sigma_{\text{basket}}/\sigma_{\text{index}}$ is then estimated from historical vol time series and used to hedge with index options.

10 Worst-of and Best-of Options**10.1 Dispersion: The Central Concept**

Dispersion measures how far asset returns spread apart. High dispersion means assets move differently from each other; low dispersion means they move together.

Dispersion and Correlation

$$\sigma_{\text{basket}}^2 = \sum_{i,j} w_i w_j \sigma_i \sigma_j \rho_{ij}$$

Lower correlation $\rho_{ij} \Rightarrow$ lower basket vol $\Rightarrow$ higher dispersion (returns spread further apart).
Therefore:

Long dispersion $\equiv$ Short correlation

Other dispersion measures: the variance-based measure above is the standard one (it links directly to variance swaps and the Stulz formula), but two alternative measures are sometimes used for reporting or robustness:

Measure	Formula
Average squared deviation	$D = \frac{1}{n} \sum_i \mathbb{E}[(R_i - \bar{R})^2], \bar{R} = \frac{1}{n} \sum_i R_i$
Average absolute deviation	$D_{\text{abs}} = \frac{1}{n} \sum_i \mathbb{E}[R_i - \bar{R}]$

The squared measure decomposes exactly into $\frac{1}{n} \sum_i \sigma_i^2 - \sigma_{\text{basket}}^2$ (up to a $1/n$ scaling), tying it directly to implied correlation and variance-swap replication — this is why it is preferred for pricing and Greeks. The absolute measure is more robust to outliers and more intuitive for risk reporting (“% average spread”), but does not decompose cleanly into variance/correlation terms and is non-differentiable at zero.

10.2 Useful Identities

The single most useful model-free identity for two assets links worst-of, best-of, and the basket:

Min–Max Identity

$$\min(S_1, S_2) + \max(S_1, S_2) = S_1 + S_2$$

This holds path-by-path, with no model assumption. Consequences:

$$\text{WO} + \text{BO} = \text{Basket (equal-weighted)}, \quad \frac{\partial \text{BO}}{\partial \rho} = -\frac{\partial \text{WO}}{\partial \rho}, \quad \Gamma_{12}^{\text{BO}} = -\Gamma_{12}^{\text{WO}}$$

Call decomposition: applying the identity to call payoffs,

$$\text{BO Call}(K) = \text{Call}(S_1, K) + \text{Call}(S_2, K) - \text{WO Call}(K)$$

This is the standard way to hedge a short BO call: **buy** the two individual calls (this over-replicates the BO call, since $\text{BO} \leq \text{Call}_1 + \text{Call}_2$), then buy back the WO call if available, or accept the over-hedge. At $\rho = -1$, the two individual calls are never simultaneously ITM, so $\text{BO Call} = \text{Call}_1 + \text{Call}_2$ exactly — this gives the *maximum* possible BO call price (and the maximum cost of a static hedge).

Put decomposition via digitals: a WO put with strike K can similarly be expressed as an integral (or, in discrete form, a strip) of WO digital puts struck across all levels below K :

$$\text{WO Put}(K) = \int_0^K \text{WO Digital Put}(k) dk$$

This mirrors the standard vanilla replication $P(K) = \int_0^K \text{Digital Put}(k) dk$, but each WO digital itself requires the joint (bivariate) distribution — it cannot be priced from individual marginal skews alone (see below).

Caution / Common Mistake

A static arbitrage exercise: selling two units of $C = \min(A, B)$ for \$100 each and using the \$200 proceeds to buy one A and one B produces a terminal payoff $A_T + B_T - 2 \min(A_T, B_T) = |A_T - B_T| \geq 0$ at zero net cost. This is a pure arbitrage bound: $\min(A, B) \leq \frac{1}{2}(A + B)$ always, regardless of any model, vol, or correlation assumption.

10.3 Worst-of Call**Worst-of Call Payoff**

$$\text{WO Call} = \max \left[0, \min_i \left(\frac{S_i(T)}{S_i(0)} \right) - K \right]$$

A worst-of call is *cheaper* than a basket call, because the payoff depends on the worst performer (lower expected return).

For two assets (Stulz formula under GBM):

$$\begin{aligned} \text{WO Call} = & S_1 N_2(\gamma_1 + \sigma_1 \sqrt{T}, a_1; \rho_1^*) + S_2 N_2(\gamma_2 + \sigma_2 \sqrt{T}, a_2; \rho_2^*) \\ & - K e^{-rT} N_2(\gamma_1, \gamma_2; \rho_{12}) \end{aligned}$$

where N_2 is the bivariate normal CDF and ρ^* are adjusted correlations.

Vega of a WO call: Two opposing effects:

1. Higher vol $\rightarrow$ higher call value (normal call effect: positive).
2. Higher vol $\rightarrow$ higher dispersion $\rightarrow$ worse worst-of return $\rightarrow$ cheaper (negative).

The net effect *depends on the level of correlation and moneyness*. In low-correlation environments, the dispersion effect dominates and the WO call seller can be long Vega on the worst-performing underlying while being short on others.

10.4 Worst-of Put

Worst-of Put Payoff

$$\text{WO Put} = \max \left[0, K - \min_i \left(\frac{S_i(T)}{S_i(0)} \right) \right]$$

A worst-of put is *more expensive* than a vanilla basket put, because it pays on the worst performer (higher expected loss).

Key risk: WO put seller is **short dispersion = long correlation**. Higher dispersion (lower correlation) → worst performer falls further → put worth more. The seller is long Vega on all underlyings (higher vol → more dispersion → worse WO).

Parameter	WO Call Seller	WO Put Buyer
Interest rates	Short	Short
Dividends	Long	Long
Borrow costs	Long	Long
Volatility	Depends (often long)	Long
Correlation	Short	Short
Skew	Depends	Long

Both the WO call seller and WO put buyer are **short correlation** (long dispersion). This is the defining feature of dispersion products.

10.5 Pricing Methodology: Why Worst-of Options Need the Full Skew

Key Intuition

Although a WO/BO payoff depends only on *terminal* levels $S_i(T)$ (no path-dependence for a European WO/BO), pricing it correctly still requires the *entire* implied volatility skew of each underlying — not just the vol at strike K . The reason: the payoff is sensitive to two distinct things at once.

1. **Where each asset finishes.** The probability that asset i finishes deep below K (not just below K) determines *how badly* it drags the worst-of down. This requires the full downside skew of asset i , since deep-OTM strikes typically carry higher implied vol than the strike K itself.
2. **How assets finish relative to each other.** Whether asset i or asset j ends up being the worst-of is itself a random outcome depending on the joint distribution of $(S_i(T), S_j(T))$. A flat-vol (single-point) model for each asset misprices this relative-ranking probability, because the true marginal distribution of each asset is not log-normal.

Modelling consequence: calibrate a **local volatility model to each underlying's own implied vol skew**, using **exact-date fitting** at the option's maturity T (not a smooth multi-maturity fit). Exact-date fitting is sufficient here — unlike for American barriers or autocalls — because the payoff is purely a function of returns at T ; there is no need for correct *forward* dynamics between today and T , only the correct *terminal* distribution at T . The n calibrated marginal local-vol processes are then simulated jointly via a correlation matrix (Cholesky) or copula.

Skew Effect on a WO Put

A higher (steeper) skew on the downside raises the implied vol used to price the region where the WO put pays off. Since this region requires *lower* prices (the put is ITM when the worst-of is low), a steeper skew $\Rightarrow$ higher probability mass priced into that low region $\Rightarrow$ **higher WO put price**. The WO put buyer is therefore **long skew** on every underlying.

For the WO call, the same steeper skew raises the chance that *some* asset collapses and drags the minimum down, which is unambiguously bad for the call (which wants the minimum to stay high). Hence the **WO call is short skew** — symmetric to its short-dispersion position established above.

10.6 Detailed Greeks: Delta, Cross-Gamma, Vega, Correlation

This subsection traces how each Greek of a two-asset worst-of evolves as the assets diverge in relative performance — the central risk-management question on any WO/BO book.

Delta. Each individual Delta Δ_i^{WO} is diluted relative to the vanilla Delta of asset i , because the payoff only reacts to asset i when it is (or is close to being) the worst performer:

$$\Delta_i^{WO} \in [\Delta_i^{\text{vanilla}}, 0] \quad (\text{for a put})$$

At equal performance, the total Delta is shared roughly equally across assets. As one asset clearly underperforms, its Delta converges to the full vanilla Delta while the other's Delta decays to zero. The *sum* of the individual Deltas is always bounded exactly as a vanilla Delta is ($\in [-1, 0]$ for a put), because under a parallel shift of all spots, the worst-of moves one-for-one — the structure behaves like a vanilla put on the worst-of level itself.

Migration of WO Put Delta as Assets Diverge ($K = 100\%$, ATM WO)

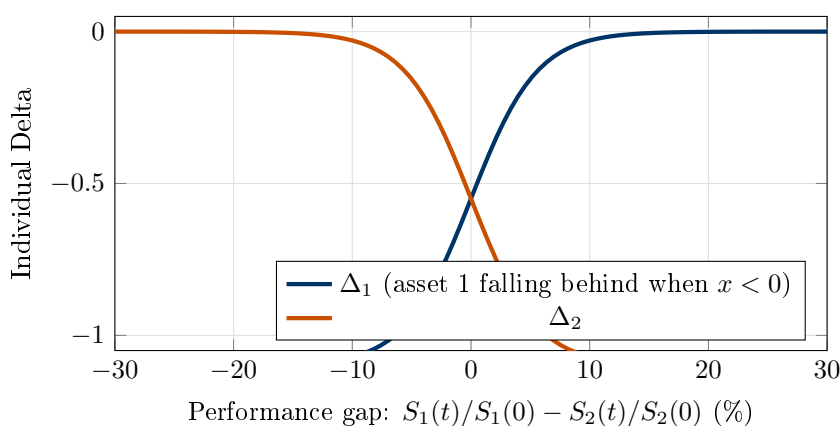


Figure 6: When assets are at parity ($x = 0$), Delta is split roughly evenly (≈ -0.5 each). As asset 2 underperforms (negative x means asset 1 ahead, asset 2 behind), its Delta migrates toward -1 (vanilla put) while asset 1's Delta decays toward 0.

Cross-Gamma. The cross-gamma $\Gamma_{12} = \partial^2 \text{WO} / \partial S_1 \partial S_2$ is **positive** for a WO option: if S_2 rises, it becomes less likely to be the worst-of, so the Delta attributed to S_1 rises (it is now more likely to be the relevant asset). By the min-max identity, $\Gamma_{12}^{BO} = -\Gamma_{12}^{WO} < 0$.

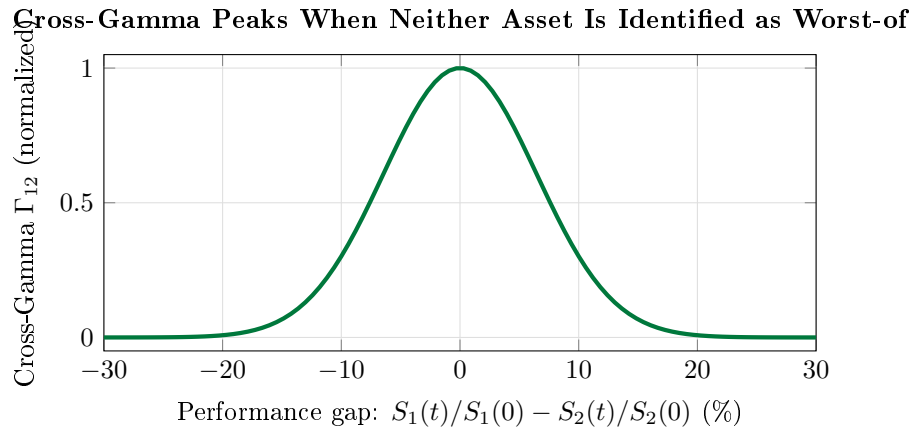


Figure 7: Cross-gamma is maximal at parity (maximum uncertainty over which asset will be worst-of) and collapses toward zero once one asset is clearly identified — at that point the structure behaves like a single-asset vanilla, which has no cross-gamma.

Vega. As established, the WO call’s Vega per asset is the sum of a positive “level” effect and a negative “dispersion” effect, with the dispersion effect dominant near parity and the level effect dominant once an asset is clearly identified as worst-of (converging to the ordinary positive vanilla Vega). For the **WO put**, both effects point the *same* way: higher vol on any asset raises the chance it drags the minimum down, which is favourable for a put. Hence **the WO put’s Vega is unambiguously positive on every underlying**, with no sign ambiguity — unlike the WO call.

Correlation Delta. $\partial \text{WO} / \partial \rho$ is maximal in the intermediate-correlation, near-parity region (where a marginal change in ρ has the largest effect on which asset is likely to end up worst), and decays to zero both as $\rho \rightarrow 1$ (assets indistinguishable, correlation no longer matters) and as one asset becomes clearly identified as the worst-of regardless of ρ (next subsection explains the mechanism in detail).

Region	Cross-Gamma	Corr. Delta	Vega per asset
Assets near parity	High (peak)	High (peak)	Split, sign ambiguous (call)
One asset emerging as WO	Falling	Falling	Migrating to the emerging asset
WO fully identified	$\rightarrow 0$	$\rightarrow 0$	$\rightarrow$ vanilla Vega (always +) on that asset

Key Intuition

All three “multi-asset” Greeks — cross-gamma, correlation Delta, and the dispersion-driven part of Vega — share the same qualitative life cycle: they are largest when there is maximum uncertainty about which asset will end up worst-of, and they collapse to zero once that uncertainty resolves. This is the formal sense in which a worst-of’s “correlation risk” is non-replicable and concentrated in the transition phase of the trade.

10.7 Correlation Transmits via the Forward (Delta), Not via Vega

For a **basket** option, correlation enters through the basket’s *volatility* (Section 9) — a Vega-type exposure. For a **worst-of** option, correlation instead acts on the **forward (expected level) of the worst-of itself** — a Delta-type exposure. This distinction is easy to miss but central to how a WO book is hedged.

Key Intuition

Why the WO forward depends on correlation. Consider n independent assets ($\rho = 0$), each equally likely to go up or down. Out of 2^n equally likely sign combinations, only 1 has *all* assets going up; the other $2^n - 1$ have *at least one* asset going down. With $n = 3$: 7 out of 8 scenarios feature at least one underperformer. Selecting the worst-of with hindsight therefore introduces a strong downward selection bias on its expected value:

$$\mathbb{E}[\min_i S_i(T)] \ll \mathbb{E}[S_i(T)] \quad \text{for any individual } i, \text{ when } \rho \text{ is low.}$$

At $\rho = 1$, there is no such bias (every asset moves together, scaled by its own vol) — the worst-of forward equals the forward of whichever asset has the highest volatility. As ρ increases from 0 to 1, the selection bias shrinks and the worst-of forward rises toward the individual forwards.

Correlation's Channel of Action on a Worst-of

$$\frac{\partial \text{Forward}_{WO}}{\partial \rho} > 0 \quad (\text{forward rises with correlation}), \quad \frac{\partial \sigma_{WO}}{\partial \rho} \approx \text{secondary}$$

A WO put can therefore be analysed like a vanilla put on a single synthetic underlying (“the worst-of”) whose **forward** is correlation-dependent. Correlation risk is consequently concentrated where the put’s *Delta* (sensitivity to its own forward) is largest in magnitude — i.e. **moderately-to-deeply ITM** — not at the strike, where Gamma/Vega of a plain vanilla would normally peak.

Practical implication: correlation risk on a WO put peaks when the worst-of is already comfortably below the strike (a real, non-marginal worst-of has emerged), not when $\min_i S_i(T)$ is hovering exactly at K . This is consistent with Section 9’s result that basket correlation risk transmits via Vega (peaking near ATM, like any Vega profile) — the WO mechanism is structurally different.

10.8 Dynamic Vega Shift: The BofA Contact’s Key Insight

Key Intuition

At inception of a worst-of trade on SPX and EuroStoxx, the Vega is roughly equally distributed across both underlyings (since neither has yet underperformed the other).

As time passes and one underlying falls significantly: say SPX becomes the clear worst-of. The payoff then approximates a single-asset put on SPX alone. The Vega migrates *entirely* to SPX: you become 100% long SPX vol, with near-zero EuroStoxx vol sensitivity.

This dynamic redistribution of Vega is a characteristic of all worst-of products. It is non-replicable with static positions in individual-asset options and must be managed dynamically.

Formally: as $\min_i S_i(T) \rightarrow S_k(T)$ (asset k clearly the worst), the worst-of put approaches a vanilla put on S_k , and the Greeks of the structure converge to those of this vanilla put. The correlation risk disappears (you only care about k ’s vol), but the volatility risk concentrates entirely on k .

11 Worst-of Autocallable Structures

11.1 Payoff Description

The most common product at BofA Paris/Zurich. A worst-of autocall pays a coupon if *all* underlyings are above the coupon trigger on an observation date (i.e. the worst return exceeds the

trigger). If the structure reaches maturity, the investor loses capital only if the worst performer has knocked through a DI barrier.

Worst-of Autocall Payoff

At each observation date t_i :

$$\text{Coupon}(t_i) = \text{Notional} \times (1 + i \cdot C) \times \mathbf{1}_{\{W(t_i) \geq H_{\text{coupon}}\}} \times \mathbf{1}_{\{\text{not previously autocalled}\}}$$

where $W(t) = \min_i [S_i(t)/S_i(0)]$ is the worst-of return.

At maturity T (if not autocalled):

If $W(T) \geq H_{\text{KI}}$: Notional $\times 100\%$

If $W(T) < H_{\text{KI}}$: Physical delivery of worst stock at strike

Snowball effect: Coupons can be *memory* coupons: if no coupon paid in year 1, year 2 coupon $= 2C$, etc. This increases the digital size in later years (more risk for the seller).

11.2 Risk Analysis

The bank (seller of the WO autocall, which buys it back as a hedge):

Risk Factor	Bank (seller of structure)	Reason
Forwards (all underlyings)	Short	Higher forward $\rightarrow$ WO higher $\rightarrow$ autocalls faster
Volatility	Long	Seller long vol through dispersion effect
Correlation	Short	Lower corr $\rightarrow$ higher dispersion $\rightarrow$ worse WO
Skew	Short (net, usually)	Short the WO digitals (long skew on DI put, but less)
Trigger levels	Long	Higher trigger $\rightarrow$ longer-lived note

BofA Context

Single stocks vs. indices:

BofA France: indices (EuroStoxx, SPX, Nikkei). Very long dated (10Y). Indices are diversified, so idiosyncratic events (takeovers, earnings, etc.) are averaged out. Long-dated index autocalls are manageable.

BofA Switzerland: single stocks. Shorter dated (2–5Y). Why shorter? Because single stocks have *idiosyncratic risk* that cannot be hedged. The example given: LinkedIn (3-year autocall). Microsoft acquired LinkedIn at a price above the coupon barrier but below the autocall barrier. Consequence: coupon keeps being paid in perpetuity, vol goes to zero (no stock movement anymore), impossible to exit. This kind of event is catastrophic on a 10-year note — hence the preference for shorter maturities on single stocks.

11.3 The Non-Replicable Correlation Problem

Key Intuition

Consider a WO autocall on SPX and EuroStoxx. When SPX falls, you become *more long SPX vol* (SPX may become the worst-of) and *less long EuroStoxx vol*. This *shift in Vega between assets as their relative performance changes* is the non-replicable risk: you cannot hedge it with a static portfolio of single-asset options.

What the model must do: it must be *statistically realistic* about the joint behavior of the assets. Even if the Vega shift cannot be replicated, the model must correctly capture the probability of each scenario (which asset becomes worst-of, and under what conditions). A model that is statistically accurate will at least allow correct pricing and correct dynamic margins for this unhedgeable risk.

Mathematically: the correlation Delta (sensitivity of the WO autocall price to a change in ρ_{ij}) is not constant. As the worst-of emerges, the structure's price becomes less sensitive to the correlation and more sensitive to the single-asset volatility of the worst performer. The correlation risk is dynamic and path-dependent, and cannot be fully hedged with correlation swaps (which are also illiquid).

12 Named Autocall Structures: Athena and Phoenix

Athena and Phoenix are the two dominant commercial autocall architectures in the French and Swiss structured products market. Both are marketed as notes (capital at risk) with high above-market coupons. They share the same fundamental autocall mechanism but differ critically in their coupon payment structure: the Phoenix pays coupons independently of the autocall trigger, while the Athena ties coupon payment entirely to the autocall event.

12.1 Athena (Knock-Out Coupon Autocall)

12.1.1 Payoff Mechanism

In an Athena, the coupon trigger and the autocall trigger are **the same level** (typically 100% of initial spot for a standard structure). There is therefore **at most one coupon paid** during the life of the note: the coupon is paid simultaneously with the autocall event.

Athena Payoff

At each annual observation date t_i ($i = 1, \dots, n$), if not previously autocalled:

$$\text{Payment}(t_i) = \text{Notional} \times (1 + i \cdot C) \times \mathbf{1}_{\{W(t_i) \geq H\}}$$

where H is both the autocall and coupon trigger (e.g. $H = 100\%$), C is the annual coupon rate (e.g. 8–12%), and $W(t_i) = \min_k S_k(t_i)/S_k(0)$ for worst-of structures.

At maturity T (if never autocalled):

$$W(T) \geq H_{\text{KI}} : \quad \text{Full capital (100\%)}$$

$$W(T) < H_{\text{KI}} : \quad \text{Physical delivery at loss (capital at risk)}$$

The **memory / snowball** feature is systematically embedded: if the note has not autocalled at t_1 , then at t_2 the coupon is $2C$; at t_3 it is $3C$, and so on. This compensates the investor for coupons missed in prior years. For the bank, this means the digital size grows with each observation: the year-3 digital pays $3C$ but has a lower conditional probability of striking.

12.1.2 Structure Example

Parameter	Typical value (French assurance-vie)
Underlying	EuroStoxx 50 or worst-of (SX5E / SPX / NKY)
Maturity	10 years
Autocall trigger H	100%
Coupon trigger	100% (same as autocall)
Annual coupon C	8–12% (memory: iC at year i)
DI put barrier	60–70%
Capital protection	No (DI put at maturity)

12.1.3 Risk Profile for the Bank

- **Short forward:** Bank benefits from a market that stays flat or mildly negative (note does not autocall, no coupon outflows). Flat market $\rightarrow$ long-lived note $\rightarrow$ bank collects the spread on the initial hedges.
- **Long Vega / short correlation:** Standard worst-of autocall risk (Section 9).
- **Snowball digital risk:** In year 8, the bank may owe $8C = 80\text{--}96\%$ in one payment if the trigger is hit. This large digital creates outsized skew and Vega exposure at that specific maturity. Pricing must be very careful about the skew at long tenors.
- **DI put at maturity:** The 10-year DI put is the hardest component to hedge. 10-year vanilla puts are illiquid; the bank must dynamically manage the put via Delta and Gamma hedges over a decade. Long dated dividends and long-dated vol uncertainty are key risks.

Interview Question / Key Insight

Q: In an Athena with memory coupons, why is the year-8 digital the riskiest for the bank?

The year-8 digital pays $8 \times C$ but only triggers if the note has survived 7 years without autocalling (i.e. the worst-of has been below 100% for 7 years). The conditional probability is very low, but the potential payout is enormous ($\approx 80\%$ of notional). The bank is effectively short a large, low-probability digital. If the market recovers sharply in year 8 after prolonged underperformance, the bank faces a catastrophic coupon payment. This is the *teeny effect*: a very cheap option (low probability) with very large payout — never ignore it.

12.2 Phoenix (Knock-In Coupon Autocall)

12.2.1 Key Difference from Athena

In a Phoenix, the coupon barrier B and the autocall barrier H are **separated**: $B < H$, typically $B \approx 70\text{--}80\%$ and $H = 100\%$. This means:

- A coupon is paid if the worst-of is above B (even if below H , so no autocall).
- The autocall only triggers if the worst-of is above H .

Consequently, the Phoenix can pay **multiple coupons without ever autocalling**. This is a fundamentally different cash flow profile from the Athena.

Phoenix Payoff

At each observation date t_i ($i = 1, \dots, n$), if not previously autocalled:

$$\text{Coupon}(t_i) = C \times \text{Notional} \times \mathbf{1}_{\{W(t_i) \geq B\}}$$

$$\text{Autocall}(t_i) = \text{Notional} \times \mathbf{1}_{\{W(t_i) \geq H\}} \quad (H > B)$$

There is **no memory**: each coupon is independent. The investor receives C whenever $W \geq B$, regardless of prior periods. At maturity, same DI put feature as the Athena.

12.2.2 Structure Example

Parameter	Typical value
Underlying	Worst-of (SX5E / SPX / NKY) or single indices
Maturity	5–8 years
Autocall trigger H	100%
Coupon barrier B	70–80%
Quarterly/Annual coupon	2–4% per period
DI put barrier	50–65%
Capital protection	No

12.2.3 Investor Appeal

The Phoenix is popular because the investor receives coupon payments even when the market is moderately down (as long as worst-of $> B$). This creates a more regular income stream and avoids the all-or-nothing profile of the Athena.

12.2.4 Risk Profile for the Bank: Athena vs. Phoenix

Risk Dimension	Athena	Phoenix
Coupon structure	Memory, paid only at autocall	No memory, paid every period if $W > B$
Digital payoff at t_i	Large (up to $n \times C$) late-life	Small (C) every period, n digitals
Autocall trigger	H (single level)	H (same)
Coupon trigger	$H = \text{autocall trigger}$	$B < H$ (separate)
Skew exposure	More concentrated (large late digits)	Spread across many small digitals
Vega position	Long (worst-of effect dominates)	Long (same logic)
Correlation	Short	Short
Complexity	High (memory)	Lower (no memory, but more coupons)

Key Intuition

Key pricing difference: In the Phoenix, each coupon is a European digital struck at B , conditional only on the note being alive (no previous autocall). Since $B < H$, the coupon digital is *in the money* relative to the autocall trigger: coupons are paid more frequently. This is what allows the Phoenix to advertise regular income. The cost of this income comes through a lower per-period coupon rate (vs. the Athena's potentially large memory coupon) and/or a lower DI put barrier.

In the Athena, the coupon and autocall are the same event. The only way the investor sees any money before maturity is if the note autocalls. This makes the Athena's cash flow *binary at each date*: either the note dies (and coupon is paid) or it lives (and nothing is paid).

BofA Context

At BofA Paris, Athena structures on EuroStoxx / SPX / Nikkei worst-of are the most common product for assurance-vie. The 10-year maturity is driven by tax law. The Phoenix appears less frequently in France (memory coupon is simpler to market) but is common in Switzerland for shorter-dated products where regular income is valued by wealth management clients.

13 Correlation: Theory and Trading**13.1 Realized and Implied Correlation****Realized Basket Correlation**

$$\rho_{\text{basket}} = \frac{\sum_{i < j} w_i w_j \sigma_i \sigma_j \rho_{ij}}{\sum_{i < j} w_i w_j \sigma_i \sigma_j}$$

This is the weighted average pairwise correlation of the basket constituents.

Implied correlation: Extracted by comparing:

- ATM straddle on the index (implies σ_{index}).
- ATM straddles on the individual constituents (implies σ_i for each).

Using $\sigma_{\text{index}}^2 = \sum_{i,j} w_i w_j \sigma_i \sigma_j \rho_{ij}$, one solves for the average implied correlation.

Correlation skew: During market crises (large downward moves), correlations spike toward 1 as stocks crash together. This creates a negative relationship between equity levels and correlation (correlation skew, analogous to the vol skew).

13.2 Correlation Matrix and Positive Definiteness

For n assets, the correlation matrix M_ρ must be:

- Symmetric: $\rho_{ij} = \rho_{ji}$.
- Diagonal entries = 1: every asset correlates perfectly with itself.
- **Positive semi-definite:** all eigenvalues ≥ 0 .

When building correlations from historical data or setting them manually, the matrix may violate positive definiteness. It must be adjusted (e.g. Higham's nearest PD matrix).

13.3 Multi-Asset Greeks**Cross-Gamma and Correlation Delta**

$$\Gamma_{ij} = \frac{\partial^2 V}{\partial S_i \partial S_j} \quad (\text{Cross-Gamma})$$

$$\Delta_\rho = \frac{\partial V}{\partial \rho_{ij}} \quad (\text{Correlation Delta})$$

Cross-Gamma measures how the Delta on S_j changes when S_i moves. It is particularly important in multi-asset structures where the worst-of may switch between assets.

Correlation Delta (the analog of Vega for correlation) measures how sensitive the structure price is to a change in pairwise correlation. For a worst-of autocall seller (short correlation): a 1% increase in correlation makes the structure cheaper, generating a gain.

13.4 Trading Correlation

Because the seller of multi-asset products is typically *short correlation*, the exotic desk structurally accumulates long-correlation positions. This can be partially hedged by:

1. **Index vs. constituent straddles:** Sell index straddles (short vol), buy straddles on constituents (long vol). This is a correlation trade: profit if realized correlation is high (index vol > constituent-weighted vol).
2. **Correlation swaps:** Pay fixed correlation (a strike), receive realized correlation. Less liquid but directly expresses correlation views. Replication is difficult because the payoff is in correlation, not variance.
3. **Variance dispersion:** Trade variance swaps on the index vs. constituents. More liquid than correlation swaps and can be replicated with strips of options.

Interview Question / Key Insight

Q: You sell a worst-of autocall on SPX/SX5E/NKY. What is your main unhedgeable risk?

The correlation shift. As one index underperforms, the structure's risk concentrates on that index alone. You cannot replicate the dynamic Vega migration with static single-asset options. The desk manages this by holding a dynamic margin (reserve capital) for this unhedged risk, sized by stress-testing the model across plausible correlation scenarios.

14 Multi-Asset Modelling

14.1 From Single-Asset to Multi-Asset Local Vol

For a multi-asset worst-of product, one needs:

1. A **local vol model for each underlying**, calibrated to its individual skew surface.
2. A **correlation matrix** linking the Brownian motions of the different assets.

Multi-Asset Local Vol SDE

$$dS_i(t) = r S_i(t) dt + \sigma_i(S_i(t), t) S_i(t) dW_i^t$$

$$\langle dW_i, dW_j \rangle = \rho_{ij} dt$$

This is the workhorse model for multi-asset exotic options at most banks. It calibrates correctly to individual skews and uses a historical or implied correlation matrix.

Calibration procedure for worst-of products:

- If the payoff depends only on terminal values (European-style WO put): use *exact date fitting* for each underlying at the option maturity.
- If path-dependent (WO autocall with daily-monitored DI barrier): use *smooth surface calibration* across all maturities for each underlying.
- Set correlations, verify positive definiteness, and run Monte Carlo simulation.

14.2 The Non-Replicable Part and Statistical Realism

As discussed in the context of the worst-of autocall: the Vega migration between underlyings as the worst-of emerges is a risk that *cannot be hedged with available instruments*.

Key Intuition

For replicable risks (Vega on each individual index), the model must be calibrated correctly (to individual skews) so that it shows risk against the hedge instruments.

For *non-replicable* risks (the correlation between assets, the migration of Vega), the model must be *statistically realistic*: it must correctly simulate the joint behavior of the assets under various market scenarios. A model that is statistically accurate allows the desk to correctly size the dynamic margin (reserve) for the unhedgeable risk.

This is why the choice of correlation matrix is critical — not just for pricing, but for risk management of the unhedgeable part of the book.

14.3 Copulas: Flexible Dependency Modelling

For hybrid products (equity + rates, equity + FX) where the marginals may have very different distributions, copulas provide a flexible way to model joint behavior.

Sklar's Theorem and the Copula

$$J_{X,Y}(x,y) = C(F_X(x), F_Y(y))$$

Any joint distribution can be written as a copula C applied to the marginal CDFs. The copula and the marginals are specified separately, allowing independent calibration.

Gaussian copula:

$$C(u,v) = N_2(N^{-1}(u), N^{-1}(v); \rho)$$

Popular due to tractability. The correlation ρ is the only free parameter. Drawback: symmetric tail dependency (no greater dependence in the tails), which is unrealistic for equity crashes (where correlations spike).

Simulation procedure (Gaussian copula + SABR marginals):

1. Simulate correlated standard Normals (ϵ_1, ϵ_2) with correlation ρ .
2. Transform to uniform: $(u, v) = (N(\epsilon_1), N(\epsilon_2))$.
3. Apply inverse SABR marginal CDFs to obtain $(S_1(T), S_2(T))$ with correct skews.

This allows calibration to individual skews (via SABR) while controlling the dependency structure (via the Gaussian copula parameter ρ).

15 Other Instruments: Cliquets, Variance Products and Cross-Asset Digitals**15.1 Cliquets (Forward-Starting Options)**

A cliquet resets its strike periodically. The simplest form: a sum of forward-starting calls, each resetting at the current spot level at the start of each period.

Basic Cliquet Payoff

$$\text{Cliquet} = \sum_{i=1}^n \max\left(F, \min\left(C, \frac{S(t_i) - S(t_{i-1})}{S(t_{i-1})}\right)\right)$$

where F = local floor, C = local cap, and optionally a global floor/cap on the total.

Key risks:

- **Forward skew:** The price depends on the skew *at each reset date in the future* (not today's skew). Local vol models predict forward skews that flatten, causing misvaluation. Stochastic vol is required.
- **Vega convexity (Volga):** Cliquets with asymmetric floors and caps have significant Volga. The Vega profile can change sign as the underlying moves. Again, stochastic vol (or careful Monte Carlo with a stochastic vol model) is needed.

15.2 Variance Swaps: Brief Overview

A variance swap delivers exposure to *realized variance* independent of the path of the underlying. The payoff at maturity is:

Variance Swap Payoff

$$\text{Payoff} = N \times (\sigma_{\text{realized}}^2 - K_{\text{var}})$$

where $\sigma_{\text{realized}}^2 = \frac{252}{N} \sum_{i=1}^N \ln^2\left(\frac{S(t_i)}{S(t_{i-1})}\right)$ and K_{var} is set to the fair variance strike at inception (zero cost to enter).

Key properties:

- Replicable with a strip of OTM options (weighted by $1/K^2$) + dynamic futures position. This makes it model-independent: the fair strike is determined by the vol surface alone.
- Constant cash Gamma: $\Gamma_{\text{cash}} = 2/T \cdot 1/S^2 \cdot S^2 = 2/T$ (constant). This means the dollar Gamma P&L is not path-dependent.
- Variance is convex in vol: a long variance swap benefits from high vol-of-vol. Long variance = long Volga. This is why variance swaps are more expensive than the square of ATM vol.
- Used to trade the implied-vs-realized vol difference (pure vol exposure without delta-hedging re-optimization).

15.3 Corridor Variance Swaps

A corridor variance swap accumulates realized variance *only on days when the spot lies within a specified corridor* $[L, U]$. Each daily squared return is multiplied by an indicator:

Corridor Variance Swap Payoff

$$\sigma_{\text{corridor}}^2 = \frac{252}{N} \sum_{i=1}^N \ln^2\left(\frac{S(t_i)}{S(t_{i-1})}\right) \cdot \mathbf{1}_{\{L \leq S(t_{i-1}) \leq U\}}$$

$$\text{Payoff} = N \times (\sigma_{\text{corridor}}^2 - K_{\text{corr}})$$

Three standard variants:

- **Full corridor** $[L, U]$: variance accrues only when the spot is inside the range. Exposure concentrated around a target level; cheaper than a standard variance swap.
- **Upside-only** $(H, +\infty)$: variance accrues only when the spot is *above* a threshold H . Isolates the vol in rallying markets; used to hedge products that suffer when the spot is high and volatile (e.g. CUO books).
- **Downside-only** $(-\infty, L)$: variance accrues only below L . Pure exposure to crash variance; natural hedge for DI put books or autocall sellers who are long downside vol.

Replication: the corridor var swap on $[L, U]$ is replicated by a strip of vanilla options restricted to strikes $K \in [L, U]$ (weighted by $1/K^2$), truncating the standard variance swap replication at the corridor boundaries. It is therefore model-independent given the vol surface in $[L, U]$, but requires liquid options across the full corridor range.

Key use case: vega bucketing by spot level — instead of a single flat Vega exposure across all spot levels, corridor var swaps let a desk isolate and hedge Vega in specific spot regions, precisely matching the region-dependent Vega profile of structured products such as autocalls (long vol below trigger, short vol above) or barrier options.

15.4 Double Digital Cross-Asset Options

A double digital cross-asset option pays a fixed coupon C if *two* underlyings simultaneously satisfy their respective conditions at maturity T :

Double Digital Cross-Asset Payoff

$$\begin{aligned}\text{Payoff} &= C \times \mathbf{1}_{\{S_1(T) \geq K_1\}} \times \mathbf{1}_{\{S_2(T) \geq K_2\}} \\ \text{Price} &= C \cdot e^{-rT} \cdot \mathbb{Q}(S_1(T) \geq K_1 \text{ and } S_2(T) \geq K_2)\end{aligned}$$

The price is the discounted *joint* probability that both conditions are met — this is the key difference from two independent single-asset digitals. Using the covariance decomposition:

$$\mathbb{Q}(A \cap B) = \mathbb{Q}(A) \cdot \mathbb{Q}(B) + \text{Cov}(\mathbf{1}_A, \mathbf{1}_B)$$

where $\text{Cov}(\mathbf{1}_A, \mathbf{1}_B) = \rho_{AB} \cdot \sqrt{\mathbb{Q}(A)(1 - \mathbb{Q}(A))} \cdot \sqrt{\mathbb{Q}(B)(1 - \mathbb{Q}(B))}$.

Correlation sensitivity: higher ρ_{12} (correlation between S_1 and S_2) $\Rightarrow$ higher joint probability $\Rightarrow$ higher price. The double digital is therefore **long correlation** — it benefits when the two underlyings are likely to satisfy their conditions simultaneously. Conversely, a seller of the double digital is **short correlation**.

Key Intuition

The double digital is the purest single-instrument expression of correlation sensitivity in multi-asset products. Its price is entirely determined by: (1) the individual digital probabilities $\mathbb{Q}(S_i \geq K_i)$ (driven by single-asset skew and vol), and (2) the joint probability, which adds the covariance term. The increment $\text{Cov}(\mathbf{1}_A, \mathbf{1}_B)$ is exactly the correlation premium — zero when $\rho = 0$ (independent), positive when $\rho > 0$ (co-movement). This mirrors the basket option where correlation enters via the basket vol (Section 9), but here the channel is the *joint probability* rather than the vol.

Variants: the two conditions can be of mixed type ($S_1 \geq K_1$ and $S_2 \leq K_2$), giving a *short* correlation payoff (the two events are more likely to co-occur when $\rho < 0$). Mixed-direction double digitals are used to express cross-asset directional views with a correlation overlay.

16 Interview Question Bank

This section collects the most commonly asked interview questions for structured products / exotic equity roles, with concise model answers.

16.1 Vanilla and Fundamentals

Interview Question / Key Insight

Q1: What is the forward price of a stock? Why does it matter for options?

$F = S_0 e^{(r-q)T}$ where q is the dividend yield. The forward is the fair delivery price in a futures contract. For options, the Black–Scholes price depends on the forward (not the spot directly): higher dividends $\rightarrow$ lower forward $\rightarrow$ lower call price. In structured products, dividends are a key risk for long-dated notes.

Interview Question / Key Insight

Q2: Why can't you have an implied vol surface with arbitrary shape?

Arbitrage constraints impose: (1) *Calendar spread* constraint: variance must be non-decreasing in T (otherwise buy a cheap long-dated call, sell an expensive short-dated one, get arbitrage). (2) *Butterfly* constraint: the PDF implied by the vol surface must be non-negative (otherwise construct a butterfly spread with negative cost and positive payoff). The vol surface must be arbitrage-free.

Interview Question / Key Insight

Q3: You sell an ATM straddle. What happens to your position if vol goes to zero?

The straddle (call + put, same strike and maturity) has $\text{Vega} = 2\mathcal{V}_{\text{call}} > 0$ for the buyer. The seller (you) is short Vega. If vol goes to zero, the straddle becomes worth 0 (no chance of expiring ITM for either leg if there is no movement). Your short straddle $\rightarrow$ you profit by the full premium received at inception. This also illustrates that Theta (time value) is entirely driven by volatility: zero vol means zero time value.

Interview Question / Key Insight

Q4: Why is an ATM call option approximately linear in volatility?

Because its Volga (second derivative w.r.t. σ) is approximately zero ATM. The price can be expanded: $C(\sigma) \approx C(0) + \mathcal{V} \cdot \sigma + \frac{1}{2} \text{Volga} \cdot \sigma^2 + \dots$. For ATM, $C(0) \approx 0$ (no intrinsic value in zero-vol world) and $\text{Volga} \approx 0$, so $C \approx \mathcal{V} \cdot \sigma$, which is linear. OTM options have positive Volga: they benefit from the convexity — that is why they command a higher implied vol (the skew).

16.2 Greeks and Risk

Interview Question / Key Insight

Q5: Explain the Gamma-Theta trade-off in detail.

From the BS PDE with delta-hedged: $\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma = rV \approx 0$ (ignoring rates for simplicity). So $\Theta \approx -\frac{1}{2}\sigma^2 S^2 \Gamma$.

Long option: $\Gamma > 0$, therefore $\Theta < 0$ (you pay time decay). You are long Gamma and short Theta. You profit if realized vol exceeds implied vol. If the stock doesn't move, you bleed daily Theta.

Short option: $\Gamma < 0$, $\Theta > 0$ (you collect time decay). You profit if realized vol is low. If the stock makes large moves, you lose on your short Gamma.

The breakeven daily move is $|\delta S| = \sigma_{\text{impl}} S / \sqrt{252}$.

Interview Question / Key Insight**Q6: How does the skew affect the Delta of a down-and-in put?**

In a world with skew, higher strike options near the barrier have higher implied vol. The Delta of the DI put is not simply $-N(-d_1)$: the skew creates a Vanna effect (Delta sensitive to vol). As the spot falls toward the barrier, vol (and skew) increases, which affects Delta. Traders often use *sticky strike* or *sticky delta* conventions to model how the skew moves with spot, and the choice significantly affects the Delta hedge.

16.3 Autocalls and Worst-of**Interview Question / Key Insight****Q7: Why are autocalls on indices more common than on single stocks for long maturities?**

Single stocks carry idiosyncratic risk: mergers, defaults, suspension of trading, regulatory events. These are impossible to model accurately over 10 years and create unhedgeable discontinuous risks. Example: a corporate action on a single stock can make its vol collapse to zero (post-acquisition), leaving the bank with unhedgeable exposure. Indices are diversified: idiosyncratic events of any constituent get averaged out.

Interview Question / Key Insight**Q8: You sell a worst-of autocall on SPX and EuroStoxx. How does your correlation exposure change as SPX underperforms?**

Initially: long vol on both, short correlation. As SPX falls significantly: SPX approaches being the clear worst-of. The payoff starts looking like a put on SPX alone. Vega migrates entirely to SPX. Correlation risk diminishes because the second asset (EuroStoxx) is far from the strike: even large EuroStoxx moves don't affect the worst-of anymore. In the limit (SPX is clearly the worst), you have zero correlation risk and 100% SPX vol risk. This shift is the non-replicable part of the worst-of autocall.

Interview Question / Key Insight**Q9: What is the effect of adding a snowball feature to a worst-of autocall?**

Snowball: if the coupon barrier is not hit at year i , the year $i + 1$ coupon becomes $2C$, year 3 becomes $3C$, etc. For the seller (bank): in later years, the digital size is larger (because the potential coupon is higher). The bank is selling a bigger digital (e.g. 30% coupon in year 3 vs. 10% in year 1) at a lower probability of striking. The combination (large digital $\times$ small probability) must be carefully managed: if the trigger is barely missed repeatedly, the bank faces a very large potential payout at year 3 or 4.

Interview Question / Key Insight

Q10: Explain why the seller of a worst-of autocall is short correlation.

The autocall pays a coupon only if *all* assets are above the trigger (i.e. the worst performer is above it). Higher correlation $\rightarrow$ all assets tend to move together $\rightarrow$ if the market is up, all are up (autocall more likely; seller stops paying coupons sooner). If the market is down, all are down together, but the WO is bounded by the worst. Net effect: higher correlation $\rightarrow$ lower dispersion $\rightarrow$ worst-of is closer to average $\rightarrow$ easier to trigger the autocall $\rightarrow$ structure terminates faster $\rightarrow$ seller collects fewer coupon exposures $\rightarrow$ structure is cheaper for the bank. Short correlation means: if correlation rises, the bank benefits. Wait — let's be precise: the seller of the WO autocall *loses* if correlation falls (assets disperse, worst-of falls further, hard to autocall, bank is stuck paying forever). The bank wants high correlation: it is **long correlation, short dispersion**.

Note: "seller of the WO autocall" = the bank that issues the note. The bank is short the structure to the client. The bank's hedging book is long correlation.

16.4 Modelling Questions

Interview Question / Key Insight

Q11: Why can't you use a local vol model to price a cliquet?

Local vol models imply that forward skew flattens as time progresses. The local vol function $\sigma(S_t, t)$ is deterministic: as you simulate forward, the model's conditional distribution of returns narrows (less uncertainty). This makes the model predict near-flat forward skews. But cliquets are directly sensitive to forward skew (they pay on forward-starting returns). A local vol model will systematically underprice cliquets. A stochastic vol model (Heston, SABR) that generates realistic forward dynamics is required.

Interview Question / Key Insight

Q12: What is the difference between exact date fitting and smooth surface calibration for local vol?

Exact date fitting: calibrate the local vol model only to the vanilla skew at the option's expiry. Used when the payoff has no path dependency (e.g. European-barrier option monitored only at maturity). Gets the terminal distribution exactly right.

Smooth surface calibration: calibrate across all maturities, ensuring realistic forward dynamics through time. Used for continuously-monitored barriers or any path-dependent product. Slightly less accurate at any individual date but ensures correct behavior at intermediate times.

17 Appendix: Key Formulas at a Glance

Black-Scholes Formula

$$C = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2)$$

$$P = Ke^{-rT}N(-d_2) - Se^{-qT}N(-d_1)$$

$$d_1 = \frac{\ln(S/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

Key Greeks

$$\begin{aligned}\Delta_C &= e^{-qT} N(d_1) & \Delta_P &= -e^{-qT} N(-d_1) \\ \Gamma &= \frac{e^{-qT} N'(d_1)}{S\sigma\sqrt{T}} \\ \mathcal{V} &= S e^{-qT} N'(d_1) \sqrt{T} \\ \Theta_C &= -\frac{S e^{-qT} N'(d_1) \sigma}{2\sqrt{T}} - r K e^{-rT} N(d_2) + q S e^{-qT} N(d_1)\end{aligned}$$

ATM Approximations (S=100%, zero rates/divs)

$$\begin{aligned}\mathcal{V}_{\text{ATM}} &\approx 0.4 S \sqrt{T} \\ C_{\text{ATM}} &\approx 0.4 \sigma S \sqrt{T} \\ \Gamma_{\text{ATM}} &\approx \frac{0.4}{S \sigma \sqrt{T}} \\ \Theta_{\text{ATM}} &\approx -\frac{0.4 S \sigma}{2\sqrt{T}} \cdot \frac{1}{252} \quad (\text{daily})\end{aligned}$$

Gamma–Theta P&L Identity

$$\text{Daily P\&L (delta-hedged)} = \frac{1}{2} \Gamma S^2 (\sigma_{\text{realized}}^2 - \sigma_{\text{implied}}^2) \Delta t$$

Breakeven daily move: $|\delta S|^* = \sigma_{\text{impl}} S / \sqrt{252}$.

Put–Call Parity

$$C - P = e^{-rT} (F - K), \quad F = S e^{(r-q)T}$$

Digital Option

$$\text{Digital Call} = e^{-rT} N(d_2), \quad \text{with skew correction: } e^{-rT} \left[N(d_2) + \mathcal{V} \left| \frac{\partial \sigma}{\partial K} \right| \right]$$

Variance Swap Payoff

$$\text{Payoff} = N \cdot (\sigma_{\text{realized}}^2 - K_{\text{var}}), \quad \sigma_{\text{realized}}^2 = \frac{252}{N} \sum_{i=1}^N \ln^2 \left(\frac{S_i}{S_{i-1}} \right)$$

Worst-of Put Payoff

$$\text{WO Put} = \max \left(0, K - \min_i \left(\frac{S_i(T)}{S_i(0)} \right) \right)$$

Barrier Shift (Broadie–Glasserman–Kou)

$$H' = H \cdot \exp \left(\pm 0.5826 \sigma \sqrt{T/m} \right)$$

*Good luck with your interview at Bank of America.
Master the intuitions — they will always ask “why” after the formula.*